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Determination of acoustic impedances of multi matching layers for narrowband ultrasonic airborne transducers at frequencies <2.5 MHz – Application of a genetic algorithm

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ABSTRACT

The effective ultrasonic energy radiation into the air of piezoelectric transducers requires using multi layer matching systems with accurately selected acoustic impedances and the thickness of particular lay ers. One major problem of ultrasonic transducers, radiating acoustic energy into air, is to find the prope acoustic impedances of one or more matching layers. This work aims at developing an original solution t the acoustic impedance mismatch between transducer and air. If the acoustic impedance defence between transducer and air be more, then finding best matching layer(s) is harder. Therefore we conside PZT (lead zirconate titanate piezo electric) transducer and air that has huge acoustic impedance defe ence. The vibration source energy (PZT), which is used to generate the incident wave, consumes a pa of the mechanical energy and converts it to an electrical one in theoretical calculation. After calculatin matching layers, we consider the energy source as layer to design a transducer. However, this part of th mechanical energy will be neglected during the mathematical work. This approximation is correct only the transducer is open-circuit. Since the possibilities of choosing material with required acoustic imped ance are limited (the counted values cannot always be realized and applied in practice) it is necessary t correct the differences between theoretical values and the possibilities of practical application of give acoustic impedances. Such a correction can be done by manipulating other parameters of matching layer (e.g. by changing their thickness). The efficiency of the energy transmission from the piezoceramic trans ducer through different layers with different thickness and different attenuation enabling a compensa tion of non-ideal real values by changing their thickness was computer analyzed (base on geneti algorithm). Firstly, three theoretical solutions were investigated. Namely, Chebyshev, Desilets and Sou quet theories. However, the obtained acoustic impedances do not necessarily correspond to a nowaday available material. Consequently, the values of the acoustic impedances are switched to the nearest val ues in a large material database. The switched values of the acoustic impedances do not generally giv efficient transmission coefficients. Therefore, we proposed, in a second step, the use of a genetic algo rithm (GA) to select the best acoustic impedances for matching layers from the material database for narrow band ultrasonic transducer that work at frequency below the 2.5 MHz by considering attenuatio However this bank is rich, the results get better. So the accuracy of the propose method increase by usin a lot of materials with exact data for acoustic impedance and their attenuation, especially in high fre quency. This yields highly more efficient transmission coefficient. In fact by using increasing numbe of layer we can increase our chance to find the best sets of materials with valuable both in acousti impedance and low attenuation. Precisely, the transmission coefficient is almost equal to unity for th all studied cases. Finally the effect of thickness on transmission coefficient is investigated for differer layers. The results showed that the transmission coefficient for air media is a function of thickness an sensitive to it even for small variation in thickness. In fact, the sensitivity increases when the difference of acoustic impedances to be high (difference between PZT and air).

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1. Introduction

Ultrasonic transducers are needed in numerous machining, forming, medical and non-destructive testing applications. However, there exist a problem of choosing an intermediat medium for impedance matching between the ultrasoni transducers and the medium to which ultrasonic energy had t be radiated. Depending on the kind of the loading mediun quarter wave matching layers are usually proposed [1]. Thi can be obtained if materials of appropriate acoustic impedanc are used.

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Although there are a lot of investigations who was concerned with piezoceramic transducers radiating a wave to liquids and biological structures (e.g. [1–4]), There are also a lot of papers were interested in the problem of matching the impedance of the ultrasonic transducer with a gas medium [5–11]. Some papers concentrate on design an barrow or broad band air born ultrasonic transducer theoretically and experimentally by using one or two matching layers and some of them concentrate on fabricate the new materials with both low acoustic impedance and attenuation [12–18]. The matching layers for air load, must be has very low acoustic impedance. There are most material such as balsa or cork wood with low acoustic impedance. But unfortunately these materials have a big attenuation in high frequency. In general we can say that materials with low acoustic impedance have high attenuation coefficient. Some researchers work on low acoustic impedance materials with low attenuation coefficient. For example they present in their investigations that the polymeric membranes are more efficient than other porous materials such as silicon rubble in high frequency. For showing the ability of the propose method, we used almost the new material properties presented by Gomez in [19]. Theoretically, it is possible to apply one, two or multilayer mechanical sets. Nevertheless, producing such layers requires high precision for transducers.As the characteristic mechanical impedance of the Lead Zirconate Titanate piezoelectric (PZT) is $Z_p = 33 \times 10^7$ kg/(m^2 s), which is highly greater than that of air ($Z_a = 427$ kg/(m^2 s)), the two mechanical impedances must be matched in order to obtain an optimal energy transmission. This consists in adding one or several intermediate matching layers. Theoretically, solutions proposed in the literature consist in using one or more quarter-wave layers on the front face of the transducer. Practically, the possibilities, of finding a material with the required acoustic impedance, are rather limited. One research direction consists in elaborating new materials with desired acoustic impedance [1–4]. Alternatively, two or more layer schemes with readily available materials, of which the impedances decrease from 33 to 0.000427 Mrayl, can be used. Increasing the number of matching layers can offer a large range of available materials but can also induce more complexity in the fabrication process. However, this approach can enlarge the range of readily available materials that can be used for the matching layers.

The objective of this paper is fourth fold. Firstly, we aim at comparing the analytical solutions to find the acoustic impedances of matching layers. Namely, we will compare the transmission coefficient obtained by using matching layers calculated with Chebyshev, DeSilets and Souquet theories [20,21]. Secondly, we interest in determining the loss in the transmission coefficient when the theoretical acoustic impedances are replaced by the nearest acoustic impedances of nowadays available materials. Thirdly, we will apply a genetic algorithm (Dijkstra algorithm (DA) [22]) in order to determine the optimal acoustic impedances of matching layers. The acoustic impedances will be chosen from available materials database. Therefore, the obtained acoustic impedances will correspond to readily available materials. Finally, we discussed on effect of thickness of each layer on transmission coefficient.

This paper will be organized in five sections. Following this introduction, Section 2 deals with the theoretical framework of sound transmission through multilayer systems with considering attenuation of layers. Subsequently, Section 3 is dedicated to the theoretical determination of matching layers by theoretical solutions proposed in literature. An emphasize will be given on the loss in the transmision coefficients when replacing the theoretical acoustic impedances by the nearest acoustic impedances corresponding to most available materials (not theoretical acoustic values) and propose different setup for different number of layers for a narrow band ultrasonic transducer at frequency work less than 2.5 MHz. Before concluding, Section 4 will present the main contribution of this work; namely, the application of a genetic algorithm to calculate the acoustic impedances of the matching layers for narrow band transducer that work in frequency less than 2.5 MHz.

2. Sound wave transmission through multilayer systems considering attenuation

The origins of multilayer transmission theory can be traced back to Rayleigh's "Theory of Sound" published in 1878 [23] which dealt, among other problems, with the problems of reflections and refractions across a boundary and referred to earlier works of Green and Poisson. The first modern works are due to Lindsay et al. [24] and Lindsay [25] who considered the case of compression wave transmission across an infinite alternating set of layers. In Ref. [24], the work was focused on waves normal to the layers. Ref. [25] considered the case of oblique incident waves. Both alternating liquid/liquid and liquid/solid layers were studied. In the case of solid layers, account had to be taken of the mode conversion at the solid interface.

The transmission of elastic waves through a stratified solid medium was first considered by Thomson [26]. A matrix method was introduced to solve the transmission problem. Shaw and Bugl [27] pointed out that Thomson [26] had made the restrictive assumption that the shear modulus is constant throughout all the layers and that continuity of shear strain at the interfaces had been used rather than the appropriate shear stress. In our work, we use a form of the matrix method as first proposed by Thomson [26]. These restrictions were repeated in the theory presented by Brekhovskikh [28] and first corrected by Haskell [29]. Subsequently, experimental and theoretical studies appear in papers by Barnard et al. [30], Plona et al. [31], Folds and Loggins [32], and Fiorito and Uberall [33]. The theory presented in these references is often considered in very general terms with waves obliquely incident on a surface and in the case of a solid, with mode conversion at the boundaries. Besides, we consider transmission through a multilayer solid at normal incidence. In this case, mode conversion can be discounted.

If the stratified medium is assumed to obey the mechanics of infinite elastic media then we have wave velocity, in the ith medium reads [34]:

$$c_i = \sqrt{\frac{\lambda_i + 2G_i}{\rho_i}}, \tag{1}$$

where λ_i, G_i and ρ_i are the first Lamé coefficient, shear modulus and density, respectively.

The situation for multilayer transmission is shown in Fig. 1. A plane attenuated longitudinal sound wave is propagating from left to right (the positive x direction) through a series of n layers of material of differing specific acoustic impedance. The incident wave travels through medium 1 and undergoes a series of reflections and transmissions in the subsequent layers until a transmitted wave emerges into medium n. The acoustic impedance and length of the kth layer ($1 \leqslant k \leqslant n$) are denoted z_k^* and l_k. The nth medium has an infinite length, i.e., $l_n = \infty$.

Subsequently, the following assumptions will be considered:

(i) The reflected wave in medium n is assumed to be nonexistent as this layer is assumed of infinite length,
(ii) Even though the bounded media would produce multiple reflections and transmissions in each layer, it is sufficient to suppose that there is only one wave in each propagating direction; provided that the boundary conditions are satisfied, these waves will include all the individual components [34,35].

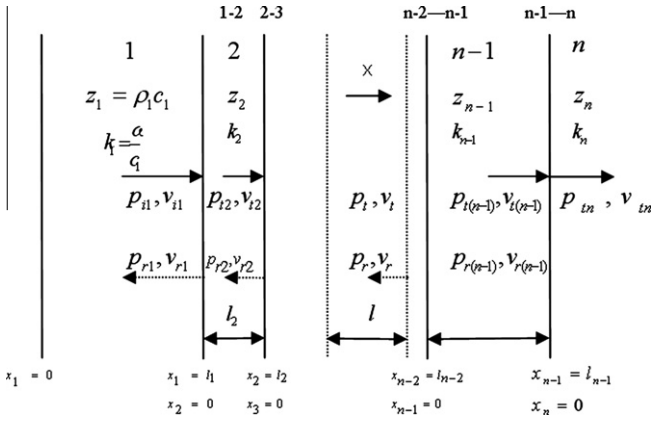


Fig. 1. A plane longitudinal sound wave transmission.

(iii) Actually, the PZT, which is used to generate the incident wave, consumes a part of the mechanical energy and converts it to an electrical one. However, this part of the mechanical energy will be neglected. This approximation is correct only if the transducer is open-circuit [33]. It was pointed out by de Klerk [36] that the great mismatch in electrical impedance, between the transducer and the amplifier, would indicate that very little energy is lost in such a manner.

(iv) We consider the wave attenuate only due to the constant loss factor (η) and this value is in depended on frequency. This assumption is logical because we calculate the loss factor from them attenuation coefficient of materials in high frequency (≈3 MHz). Therefore this is upper band estimation and if the set up work with this assumption then can work for fewer frequencies too.

Considering the first layer, we denote p_{i1} and v_{i1}, the incident wave pressure and particle velocity, respectively. Similarly, the reflected wave pressure and particle velocity are denoted p_{r1} and v_{r1}. Now let us focus on an intermediate medium, the kth medium ($1 < k < n$). There are two waves propagating in opposite directions the transmitted and reflected waves. The transmitted wave has pressure p_{tk} and particle velocity v_{tk} and the reflected wave has p_{rk} and particle velocity v_{rk}. The nth layer has only one propagating wave as it is semi-infinite, the transmitted wave, of which the pressure and particle velocity are denoted p_{tn} and v_{tn}, respectively. Assuming harmonic wave representation, the pressure and particle velocity, of the incident wave, read:

$$\begin{cases} p_{i1}(x_1) = A_1^* e^{j(\omega t - k_1^* x_1)}, & \text{(a)} \\ v_{i1}(x_1) = \frac{p_{i1}}{z_1^*} = \frac{A_1^*}{z_1^*} e^{j(\omega t - k_1^* x_1)}, & \text{(b)} \end{cases} \tag{2}$$

where A_1^* is the complex amplitude of the incident pressure wave and x_1 ($0 \leqslant x_1 \leqslant l_1$) is the longitudinal space coordinate in the first layer. $k_1^* = \frac{\omega}{c_1^*}$ is the complex wave number. Similarly, the pressure and the particle velocity of the transmitted wave in the kth layer ($1 < k \leqslant n$), are given by:

$$\begin{cases} p_{tk}(x_k) = A_k^* e^{j(\omega t - k_k^* x_k)}, & \text{(a)} \\ v_{tk}(x_k) = \frac{p_{tk}}{z_k^*} = \frac{A_k^*}{z_k^*} e^{j(\omega t - k_k^* x_k)}, & \text{(b)} \end{cases} \tag{3}$$

In Eqs. (2), A_k^* is the amplitude of the transmitted pressure wave and x_k ($0 \leqslant x_k \leqslant l_k$) is the longitudinal space coordinate. Considering the reflected wave in the kth layer ($1 \leqslant k < n$), the pressure and the particle velocity are written:

$$\begin{cases} p_{rk}(x_k) = B_k^* e^{j(\omega t + k_k^* x_k)}, & \text{(a)} \\ v_{rk}(x_k) = -\frac{p_{rk}}{z_k^*} = -\frac{B_k^*}{z_k^*} e^{j(\omega t + k_k^* x_k)}, & \text{(b)} \end{cases} \tag{4}$$

where B_k^* is the complex amplitude of the reflected pressure wave. By adding the contribution of the two waves propagating in opposite directions, the pressure and the particle velocity, in the kth medium ($1 \leqslant k < n$), are obtained:

$$\begin{cases} P_k(x_k) = A_k^* e^{j(\omega t - k_k^* x_k)} + B_k^* e^{j(\omega t + k_k^* x_k)}, & \text{(a)} \\ V_k(x_k) = \frac{A_k^*}{z_k^*} e^{j(\omega t - k_k^* x_k)} - \frac{B_k^*}{z_k^*} e^{j(\omega t + k_k^* x_k)}, & \text{(b)} \end{cases} \tag{5}$$

Eqs. (4) can be rewritten in a matrix form as

$$\begin{bmatrix} P_k(x_k) \\ V_k(x_k) \end{bmatrix} = e^{j\omega t} \begin{bmatrix} e^{-jk_k^* x_k} & e^{jk_k^* x_k} \\ \frac{1}{z_k^*} e^{-jk_k^* x_k} & -\frac{1}{z_k^*} e^{jk_k^* x_k} \end{bmatrix} \begin{bmatrix} A_k^* \\ B_k^* \end{bmatrix}. \tag{6}$$

As there is no reflection in nth layer, we can consider that:

$$B_n = 0. \tag{7}$$

Furthermore, we can assume that the amplitude of the incident pressure A_1^* is known. By considering, Eq. (5) at the two extremities for each layer, i.e., $x_k = 0$ and $x_k = l_k$ and by eliminating A_k^* and B_k^*, we obtain:

$$\begin{bmatrix} P_k(l_k) \\ V_k(l_k) \end{bmatrix} = \begin{bmatrix} e^{-jk_k^* l_k} & e^{jk_k^* l_k} \\ \frac{1}{z_k^*} e^{-jk_k^* l_k} & -\frac{1}{z_k^*} e^{jk_k^* l_k} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \frac{1}{z_k^*} & -\frac{1}{z_k^*} \end{bmatrix}^{-1} \begin{bmatrix} P_k(0) \\ V_k(0) \end{bmatrix}, \quad (1 \leqslant k < \tag{8}$$

Until now, we have only considered each layer separately. Furthermore, we can write that the pressure and particle velocity are continuous at each interface between two layers, i.e.,

$$\begin{cases} P_k(l_k) = P_{k+1}(0), & \text{a} \\ V_k(l_k) = V_{k+1}(0), & \text{b} \end{cases}, \quad (1 \leqslant k < n) \tag{9}$$

Eqs. (7) and (8) yield:

$$\begin{bmatrix} P_k(l_k) \\ V_k(l_k) \end{bmatrix} = \begin{bmatrix} e^{-jk_k^* l_k} & e^{jk_k^* l_k} \\ \frac{1}{z_k^*} e^{-jk_k^* l_k} & -\frac{1}{z_k^*} e^{jk_k^* l_k} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \frac{1}{z_k^*} & -\frac{1}{z_k^*} \end{bmatrix}^{-1} \begin{bmatrix} P_{k-1}(l_{k-1}) \\ V_{k-1}(l_{k-1}) \end{bmatrix}, \quad (1 < k < n). \tag{10}$$

Since the yang modules convert to the complex form, $E^* = E' + E''$, while considering the hysterics loss, E' the real part is the yang modulus and E'' the imaginary part that consider the loss effect. Therefore viscous loss can define by η with $\eta = \frac{E''}{E'}$ relation. We can obtain the complex wave number, k^*, and the complex acoustic impedance, z^*, in following form

$$E^* = E'(1 + \eta j)$$

$$k^* = \frac{\omega}{c^*} = \frac{\omega}{\sqrt{\frac{E^*}{\rho}}} = \frac{\omega\sqrt{\rho}}{\sqrt{E'(1+\eta j)}} = \omega\sqrt{\frac{\rho}{E'}}\frac{1}{\sqrt{(1+\eta j)}}$$

$$= \frac{k}{(1+\eta^2)^{1/4}}\left(\cos\left(\frac{1}{2}\arctan\eta\right) - j\sin\left(\frac{1}{2}\arctan\eta\right)\right) = k_1 + jk_2$$

$$k_1 = \frac{k}{(1+\eta^2)^{1/4}}\cos\left(\frac{1}{2}\arctan\eta\right)$$

$$k_2 = \frac{-k}{(1+\eta^2)^{1/4}}\sin\left(\frac{1}{2}\arctan\eta\right) \tag{10a}$$

Measuring is the Best way to calculation Attenuation for materials. Some researcher, experimentally, measured the attenuation of different low acoustic impedance at high frequencies. In this study we assume the simplified Eq. (10b) that express the relation between attenuation coefficient, α, and loss factor, η,

$$\eta = 2\zeta = -2a/\omega_n = \tan(\Phi) = E'/E'' = Q^{-1} = \lambda\alpha/\pi \tag{10b}$$

lere ζ is the relative damping, Φ is the phase difference between ension and strain, a is the decay exponent, ω_n is the frequency of ree vibrations, E' and E'' are the real and imaginary parts of the omplex Young's modulus, Q is the quality factor λ is the elastic avelength and α is the attenuation. This assumption is not so ar, because we consider the attenuation of the materials from the xperiments in high frequency (≈3 MHz) and use them to calculate oss factor. Therefore we have an estimation to consider the effect of ttenuation in our equation by using this technique.

$$\begin{aligned} z &= 1 + j\eta = re^{j\theta} \;\Rightarrow\; z^{\frac{1}{2}} = r^{\frac{1}{2}} e^{j\frac{\theta}{2}} \\ &= (1+\eta^2)^{1/4}\left(\cos\left(\tfrac{1}{2}\arctan\eta\right) + j\sin\left(\tfrac{1}{2}\arctan\eta\right)\right) \\ \theta &= \arctan\eta \\ z_i^* &= \rho_i c_i^* = \rho_i\sqrt{\frac{E_i^*}{\rho_i}} = \rho_i c_i\sqrt{1+j\eta_i} \\ &= z_i(1+\eta^2)^{1/4}\left(\cos\left(\tfrac{1}{2}\arctan\eta\right) + j\sin\left(\tfrac{1}{2}\arctan\eta\right)\right) \end{aligned} \tag{11}$$

fter evaluating a matrix inverse and a product between two matrices in Eq. (9), we get:

$$\begin{aligned} &\begin{bmatrix} P_k(l_k) \\ V_k(l_k) \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} e^{k_{2k}l_k}\times e^{-jk_{1k}l_k} + e^{-k_{2k}l_k}\times e^{jk_{1k}l_k} & +\frac{E_k^* k_k^*}{\omega}\left(e^{-k_{2k}l_k}\times e^{jk_{1k}l_k} - e^{k_{2k}l_k}\times e^{-jk_{1k}l_k}\right) \\ +\frac{\omega}{E_k^* k_k^*}\left(e^{-k_{2k}l_k}\times e^{jk_{1k}l_k} - e^{k_{2k}l_k}\times e^{-jk_{1k}l_k}\right) & e^{k_{2k}l_k}\times e^{-jk_{1k}l_k} + e^{-k_{2k}l_k}\times e^{jk_{1k}l_k} \end{bmatrix} \\ &\times \begin{bmatrix} P_{k-1}(l_{k-1}) \\ V_{k-1}(l_{k-1}) \end{bmatrix} \quad (1<k<n). \end{aligned} \tag{12}$$

We can rewrite the Eq. (10):

$$\begin{aligned} &\begin{bmatrix} P_k(l_k) \\ V_k(l_k) \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} \cos k_{1k}l_k\left(e^{k_{2k}l_k} + e^{-k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) & -\frac{z_k[k_1-\eta k_2+j(k_2+\eta k_1)]}{k_k}\left(\cos k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} + e^{k_{2k}l_k}\right)\right) \\ \frac{k_k}{z_k[k_1-\eta k_2+j(k_2+\eta k_1)]}\left(\cos k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} + e^{k_{2k}l_k}\right)\right) & \cos k_{1k}l_k\left(e^{k_{2k}l_k} + e^{-k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) \end{bmatrix} \\ &\times \begin{bmatrix} P_{k-1}(l_{k-1}) \\ V_{k-1}(l_{k-1}) \end{bmatrix} \quad (1<k<n) \end{aligned} \tag{13}$$

n order to link the pressures and the particle velocities in both the rst and last medium, we use the mathematical induction method, hich yields:

$$\begin{bmatrix} P_n(0) \\ V_n(0) \end{bmatrix} = \begin{bmatrix} P_{n-1}(l_{n-1}) \\ V_{n-1}(l_{n-1}) \end{bmatrix} = [T_{n-1}][T_{n-2}]\cdots[T_3][T_2]\begin{bmatrix} P_1(l_1) \\ V_1(l_1) \end{bmatrix} \tag{14}$$

here

$$[T_k] = \frac{1}{2}\begin{bmatrix} \cos k_{1k}l_k\left(e^{k_{2k}l_k} + e^{-k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) & -\frac{z_k[k_1-\eta k_2+j(k_2+\eta k_1)]}{k_k}\left(\cos k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} + e^{k_{2k}l_k}\right)\right) \\ -\frac{k_k}{z_k[k_1-\eta k_2+j(k_2+\eta k_1)]}\left(\cos k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} + e^{k_{2k}l_k}\right)\right) & \cos k_{1k}l_k\left(e^{k_{2k}l_k} + e^{-k_{2k}l_k}\right) + j\sin k_{1k}l_k\left(e^{-k_{2k}l_k} - e^{k_{2k}l_k}\right) \end{bmatrix}, \quad (1<k<n) \tag{15}$$

s the transfer matrix of the kth layer. By considering in Eq. (14), the xpressions of the pressure and the particle velocity given in Eq. (5), ve obtain:

$$\begin{bmatrix} 1 & 1 \\ \frac{1}{z_n^*} & -\frac{1}{z_n^*} \end{bmatrix}\begin{bmatrix} A_n^* \\ B_n^* \end{bmatrix} = [T_{n-1}][T_{n-2}]\cdots[T_3][T_2]\begin{bmatrix} e^{-jk_1^* l_1} & e^{jk_1^* l_1} \\ \frac{1}{z_1^*}e^{-jk_1^* l_1} & -\frac{1}{z_1^*}e^{jk_1^* l_1} \end{bmatrix}\begin{bmatrix} A_1^* \\ B_1^* \end{bmatrix} \tag{16}$$

or equivalently:

$$\begin{bmatrix} A_n^* \\ B_n^* \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \frac{1}{z_n^*} & -\frac{1}{z_n^*} \end{bmatrix}^{-1}[T_{n-1}][T_{n-2}]\cdots[T_3][T_2]\begin{bmatrix} e^{-jk_1^* l_1} & e^{jk_1^* l_1} \\ \frac{1}{z_1^*}e^{-jk_1^* l_1} & -\frac{1}{z_1^*}e^{jk_1^* l_1} \end{bmatrix}\begin{bmatrix} A_1^* \\ B_1^* \end{bmatrix} \tag{17}$$

As A_1^* is known and $B_k^* = 0$, Eq. (17) is a two linear equation system with two unknown variables. Hence, it can be solved to obtain A_n^* and B_1^*. Subsequently, we can deduce the sound-pressure transmission coefficient (PTC), for a system of n layers α_{pn}. This is given by

$$\alpha_{pn} = \left|\frac{A_n^*}{A_1^*}\right|. \tag{18}$$

Similarly, we can obtain the sound-power transmission coefficient α_{wn}, which reads:

$$\alpha_{wn} = \left|\frac{z_1^*}{z_n^*}\right|\left|\frac{A_n^*}{A_1^*}\right|^2. \tag{19}$$

Just note that from now to the end, in all calculation we use Eq. (19) as a transmission coefficient to calculate the efficiency of the transducer to transmission of the power.

3. Theoretical calculation of matching layers impedances

3.1. Basics

We recall that in this paper, we aim at calculating the acoustic impedance of the matching layers, i.e., $z_2, \ldots, z_{n-1}$ which give the maximum sound-pressure or sound-power transmission coefficients defined in Eq. (19). By considering a system of three layers, i.e., with only one matching medium, the Chebyshev, DeSilets and Souquet theories stipulate that the optimal transmission is obtained for [8]:

$$z_2^{(3),Ch} = \sqrt{z_1 \cdot z_3} \tag{20a}$$

$$z_2^{(3),Des} = \sqrt[3]{z_1 \cdot (z_3)^2} \tag{20b}$$

$$z_2^{(3),Sou} = \sqrt[3]{2 \cdot z_1 \cdot (z_3)^2} \tag{20c}$$

where $z_\nu^{(\mu),\chi}$ is the acoustic impedance of the νth layer of a system made of μ layers and obtained by the method χ.Eqs. (20) give the

optimal acoustic impedance. However, this optimal value may not correspond to an existing material. To avoid conceiving a new material with the required acoustic impedance, we can use multiple matching layers. The above theories can be extended to this case. Precisely, in the general case of a system of μ layers ($\mu > 3$), the optimal acoustic impedances are:

$$z_v^{(\mu),Ch} = \sqrt{z_{v-1}^{(\mu-1),Ch} \cdot z_{v+1}^{(\mu-1),Ch}}, \tag{21a}$$

$$z_v^{(\mu),Des} = \sqrt[3]{z_{v-1}^{(\mu-1),Des} \cdot (z_{v+1}^{(\mu-1),Des})^2}, \tag{21b}$$

$$z_v^{(\mu),Sou} = \sqrt[3]{2 \cdot z_{v-1}^{(\mu-1),Sou} \cdot (z_{v+1}^{(\mu-1),Sou})^2}. \tag{21c}$$

Furthermore, these theories assume that the matching layer lengths are of the form:

$$l_k = (2m-1)\pi \frac{\lambda_k}{4}, \tag{22}$$

where $\lambda_k = c_k/\omega$ is the wavelength and m is an integer number. That is why the optimal solutions are independent of the frequency. In the above theories, the acoustic impedances of the matching layers are independent of frequency. The frequency only affects the layers lengths which are calculated by using Eq. (22).

3.2. Results

To illustrate the above theories, we consider a piezo-ceramic transducer of acoustic impedance $z_1 = Z_C = 33 \times 10^6$ kg/(m^2 s). This transducer works in frequency less than 2.5 MHz should radiate into air, of which the acoustic impedance is $z_n = Z_A = 427$ kg/(m^2 s). It is worth to notice the important difference between the acoustic impedances of the two media, which will lead to almost no transmission to the air. Hence, one or several matching layer(s) is (are) needed. The length(s) of this (these) matching layer(s) will be determined by Eq. (22).

Firstly, we applied Eqs. (20) and (21) to systems made of thre to twelve layers. The obtained acoustic impedances are given i Table 1 (columns four to six). For sake of clarity, only acousti impedances for the cases of three, four and five layers are giver Subsequently, we calculate the sound power transmission coeffi cients which are depicted in Fig. 2 for both with and with ou attenuation coefficient. The Chebyshev theory gives a transmissio coefficient which is almost equal to 1 for the all studied cases whe we neglect the effect of attenuation. This means that Chebyshe theory gives good solution whatever the number of layers. O the other hand, the DeSilets yields increasing transmission coeffi cient with increasing number of layers. The transmission coeffi cient reaches the optimum value of unity for the case of seve layers. Finally, the Souquet gives increasing transmission coeffi cient with increasing number of layers for a number of layers lowe than or equal to six. The transmission coefficients obtained wit DeSilets and Souquet theories are lower than that obtained b the Chebyshev theory whatever the number of layer.

The acoustic impedances given in columns four to six of Table may not correspond to nowadays available materials. As a secon step, we build a database including the readily available material to the best knowledge of the authors. The richer bank data give the more accurate and practical results. At the end, this databas includes 297 materials and their general and acoustic properties Some of them illustrate in Table 2. Subsequently, we choose fror this database the acoustic impedances which are the nearest t the acoustic impedances obtained by Eqs. (20) and (21). For exam ple, the theoretical acoustic impedance of 118,705 kg/(m^2 s) is re placed from the database by 123,480 kg/(m^2 s). This acousti impedance corresponds to a kind of polymeric membrane calle MF-membrane filter-[15]. The new acoustic impedances are give in Table 1 (columns seven to nine). We should notice that it is har to find (fabricate as a plate) a solid material with acoustic imped ance lower than 8400 kg/(m^2 s) with low attenuation, which corre sponds to silicon soft bubble rubber. The transmission coefficien are then calculated according to the new acoustic impedance

Table 1
Acoustic impedances obtained by the three theories.

Method			Theoretical value (Eqs. (20) and (21))			Nearest from database		
			Tchebychev	Desilete	Souquet	Tchebychev	Desilets	Souquet
Acoustic impedance (kg/(m^2 s))	3 Layers	2nd Layer	118,705	18,010	22,686	123,480	17,000	17,000
	4 Layers	2nd Layer	1,979,208	217,689	319,737	1,840,000	211,000	313,000
		3rd Layer	7119	1475	2007	8400	8400	8400
	5 Layers	2nd Layer	8,081,699	1,144,627	1,862,656	7,450,000	1,100,000	1,840,000
		3rd Layer	118,701	7725	13,578	123,480	8400	110,00
		4th Layer	1743	646	902	8400	8400	8400

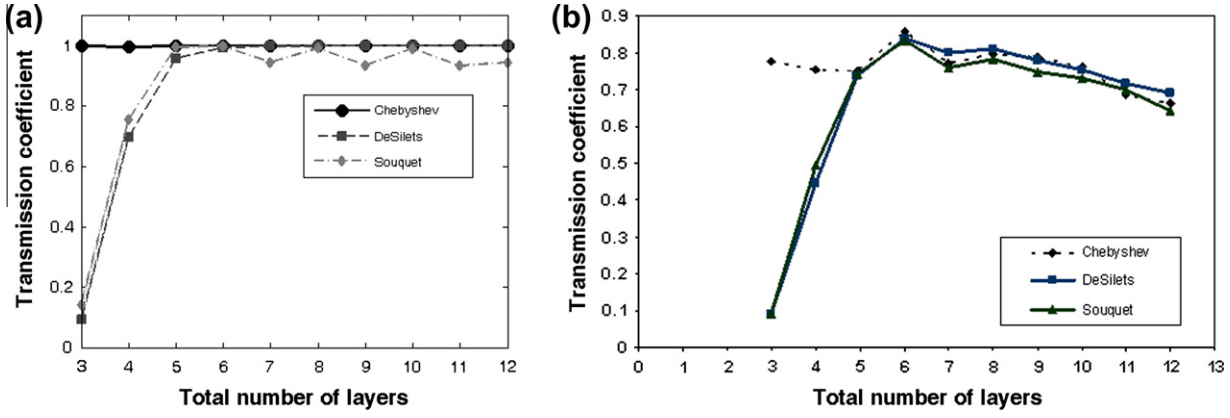


Fig. 2. Transmission coefficients obtained by the three theories. (Exact values of the acoustic impedances)

able 2
laterials and acoustic properties measured from the λ/2 thickness resonance [19].

Material	Pore size (μm)	λ/2 resonant frequency (MHz)	γ at λ/2 (NP)	Impedance (MRayl)
Vinylic/acrylic copolymer[a]	0.80	0.60	0.287	0.074
Polyethersulfone[a]	0.80	1.03	0.150	0.100
Polyethersulfone[a]	0.45	1.30	0.130	0.131
Polyethersulfone[a]	0.20	1.50	0.111	0.244
Polyethersulfone[a]	0.10	2.40	0.073	0.254
Nylon[a]	0.45	2.00	0.137	0.162
Nylon[a]	0.20	3.20	0.151	0.313
PVDF[a]	0.45	1.92	0.322	0.228
Polypropylene[a]	0.45	1.40	0.530	0.074
Polypropylene[a]	0.20	1.82	0.439	0.081
Acrylic copolymer[a]	0.30	No thickness resonances observed		
Acrylic copolymer[a]	1.20	No thickness resonances observed		
Polyethersulfone/copolymer[a]	0.45	1.82	0.250	0.203
Mixed cellulose esters[b]	5.00	0.70	0.249	0.095
Mixed cellulose esters[b]	3.00	0.86	0.198	0.094
Mixed cellulose esters[b]	1.20	0.80	0.249	0.109
Mixed cellulose esters[b]	0.80	0.75	0.212	0.083
Mixed cellulose esters[b]	0.65	1.00	0.250	0.095
Mixed cellulose esters[b]	0.45	1.00	0.216	0.098
Mixed cellulose esters[b]	0.22	1.30	0.202	0.150
Mixed cellulose esters[b]	0.10	2.40	0.190	0.256
Mixed cellulose esters[b]	0.025	4.00	0159	0.557
PVDF (hydrophobic)[b]	0.22	No thickness resonances observed		
PVDF (hydrophilic)[b]	0.22	No thickness resonances observed		
PVDF[b]	0.10	4.3	0.140	0.638
PTFE[b]	0.50	0.30	0.600	0.017
Cellulose nitrate[c]	0.65	1.03	0.246	0.084
Cellulose nitrate[c]	0.30	1.60	0.209	0.165
Cellulose nitrate[c]	0.20	1.94	0.206	0.211
Cellulose nitrate[c]	0.10	2.00	0.221	0.243

[a] Pall Corporation, Pall GelmanLaboratory ,product catalog, Ann Arbor, MI 48103–9019, http://www.pall.com.
[b] Millipore, Billerica, MA 01821, http://www.Millipore.com.
[c] Whatman International, Ltd. Whatman House, Kent ME16 oLS, U.K., http://www.whatman.com.

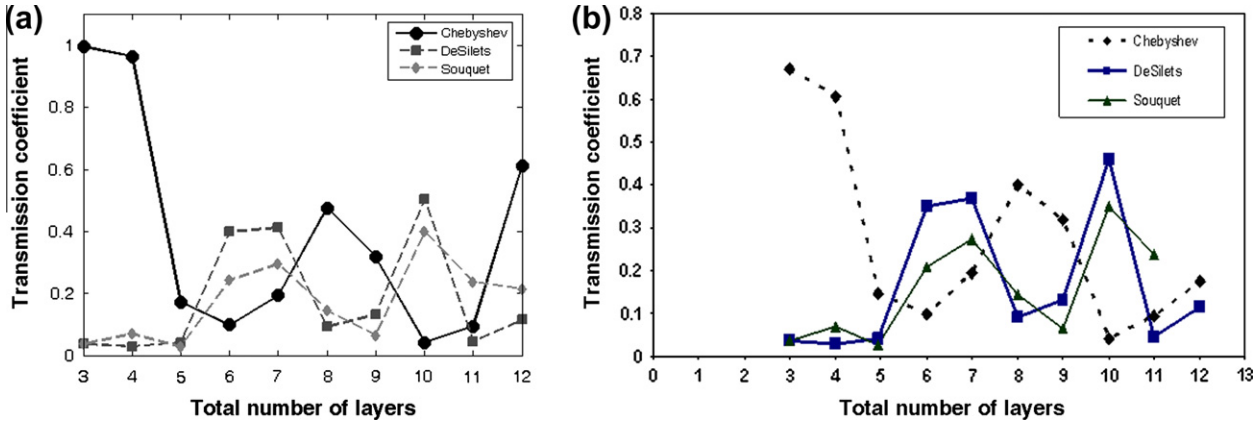


Fig. 3. Transmission coefficients obtained by the three theories (acoustic impedances values switched to nearest in a large material database).

nd depicted in Fig. 3. No clear tendency appears for the curves relative to the three theories. However, the main conclusion which ollows from Fig. 3 is that the transmission coefficients sharply rop when the theoretical acoustic impedances are switched to he nearest from the material database.

At this stage, we can make the following conclusions:

- The acoustic impedances obtained by Chebyshev, DeSilets and Souquet theories do not correspond always to nowadays available materials.
- Replacing these acoustic impedances by the nearest from a wide material database leads generally to significantly lower transmission coefficient, mainly, when the number of layers is greater than or equal to four.
- Chebyshev theory gives more efficient matching layers than the two others theories.

4. Calculation of matching layers impedances with a genetic algorithm

In the previous section, we have applied the theories of Chebyshev, Desilets and Souquet to calculate the acoustic impedances of the matching layers. Subsequently, we switch the values of these acoustic impedances to the nearest from a wide material database. This leads to a sharp drop in the transmission coefficients. The main problem of the above solutions is that the theoretical values are firstly chosen from a continuous set of possible values which is the set of strictly positive real numbers. Below, we propose a

second solution. The optimal acoustic impedances are chosen from a discrete set corresponding to the material database. Moreover, the optimal values are obtained by a genetic algorithm.

4.1. Solving constraint satisfaction problems using GA

Genetic algorithm is a computational model simulating the process of genetic selection and natural elimination in biologic evolution. Pioneering work in this field was conducted by Holland in the 1960s [37]. As a high efficient search strategy for global optimization, genetic algorithm demonstrates favorable performance on solving the combinatorial optimization problems. With comparing to traditional search algorithms, genetic algorithm is able to automatically acquire and accumulate the necessary knowledge about the search space during its search process, and self-adaptively control the entire search process through random optimization technique. Therefore, it is more likely to obtain the global optimal solution without encountering the trouble of combinatorial explosion caused by disregarding the inherent knowledge within the search space. It has been used to solve combinatorial optimization problems and non-linear problems with complicated constraints or non-differentiable objective functions. It necessitates the application of genetic algorithm into GIS route finding algorithms.

The computation of usual genetic algorithm is an iterative process that simulates the process of genetic selection and natural elimination in biologic evolution. For each iteration, candidate solutions are retained and ranked according to their quality. A fitness value is used to screen out unqualified solutions. The fitness value in this work is the transmission coefficient that calculates by the Eq. (16). In fact this value check the chromosome that selected if this fitness value is near enough to 1 that chromosome selected if not this chromosome change to another one. Genetic operators, such as crossover, mutation, translocation and inversion, are then performed on those qualified solutions to estimate new candidate solutions of the next generation. The above process is carried out repeatedly until certain convergent condition is met. And finally, GA is a general search method. It uses analogs of genetic operators on a population of states in a search space to find those states that have high fitness values. In optimization problems, genetic operators and coding methods are designed in advance so that the individuals may satisfy the constraints. In contrast, the objective of constraint satisfaction problems (CSPs) is to find an individual that satisfied constraints as the fitness value. Search in usual GA is conducted by crossover and mutation. Crossover is generally considered a robust search means. Offspring may inherit partial solutions without conflict from their parents, but no information to decide which genes are partial solutions is available. From a search strategic point of view this means that variables are randomly selected and then values which will be assigned to them are also randomly selected from genes contained within a population. Therefore search by crossover in GA cannot be considered efficient. Furthermore, mutation is a random search method itself. When we think about efficiency of global search that is the most important characteristic of GA; we must admit that the rate of search is low, because GA stresses random search rather than directional search. It can be considered that the above problem is directly inherited from problems of the neo-Darwinian evolutionary theory. In the following sections, we first explain our basic strategies. Next, we describe the fitness function, genetic operators.

4.2. Calculation of acoustic impedance according to GA theory

4.2.1. Anatomy of a genetic algorithm in Jens for the present work

A genetic algorithm (GA) can be understood as an 'intelligent' algorithm with probabilistic search. Its original idea comes from Darwinian's evolution theory. During the process of evolution, natural populations evolve according to the principles of natural selec tion and 'survival of the fittest'. Individuals who can fit in wit their surroundings more successfully will have better chances t survive and multiply, while those who do not adapt to their envi ronment will be eliminated. This means that the genes from th highly fit individuals will spread to an increasing number of indi viduals in each successive generation. The combination of goo characteristics from highly adapted ancestors may produce eve more fit offspring. In this way, species evolve to become mor and better adapted to their environment. A GA simulates thes processes by taking an initial population of individuals and apply ing genetic operators in each reproduction. In optimization terms each individual in the population is encoded into a string or chro mosome or some float number which represents a possible solu tion to a given problem. The fitness of an individual is evaluate with respect to a given objective function. Highly fit individual or solutions are given opportunities to reproduce by exchangin some of their genetic information, in a crossover procedure, wit other highly fit individuals. This produces new 'offspring' solution (i.e., children), which share some characteristics taken from bot parents. Mutation is often applied after crossover by altering som genes or perturbing float numbers. The offspring can either replac the whole population or replace less fit individuals. This evalua tion–selection–reproduction cycle is repeated until a satisfactor solution is found. The basic steps of a simple GA are show below.

Algorithm 1. (*A simple GA*)

Generate an initial population;
Evaluate fitness of individuals in the population;
repeat:
Select parents from the population;
Recombine (mate) parents to produce children;
Evaluate fitness of the children;
Replace some or all of the population by the children;
until a satisfactory solution has been found.

GAs have been developed rapidly since the past 20 years as a effective and simple optimization technique. The superiority c GAs lies in its high ability to perform parallel operations, goo robustness and convenience to realize global optimization, whic conventional optimization techniques can seldom attain. Therefor recently GAs are widely used in scientific and technical areas, suc as computer science, automation systems, electronic engineering and bio-information.

In this paper, we will focus on using GA to solve a class of opti mization problem (sound transmission from PZT transducer to ai media with finding best matching layer) as:

$$\max f(x) \text{ s.t.} x \in S, \tag{23}$$

where S is an n-dimensional box constraint, i.e., $S = \{x = (x_i)_{n\times 1} \in R^n | a_i \leqslant x_i \leqslant b_i, i = 1, \ldots, n\}$. $f(x)$ is a continuous function on S. Man models of practical problems can be concluded as the problem (23 The local optimum of (23) can be obtained by many conventiona methods, such as (quasi-)Newton method, conjugate gradien method, and trust region method. (Since there are only box con straints (23) can easily be transformed to an unconstrained prob lem.) But these methods can hardly find a global optimum whe $f(x)$ on S is complex, i.e., there are many local optimums on S. There fore, we turn to GA for its advantages in global optimization. T Algorithm 1, we usually generate an initial population randoml on S, evaluate fitness of parents and children by the target functio $f(x)$ (i.e., given an individuals x_i in S, the fitness of x_i is $f(x_i)$) and se lect parents from the population by probable rule.

In present work, each gene is a number between 1 and n (n is he number of material) and each number related to the acoustic alue of the material. For example 1 expresses the air acoustic mpedance (427 kg/(m^2 s)) and 2 present silicon bubble rubber nd so on. Each chromosome is made of m genes (m is the number f matching layer). In other meaning, each chromosome is a sort of coustic impedance of material and length of chromosome is equal o the number of layers. We need to consider an objective function o find the best solution that must maximize the transmission oefficient. It means that transmission coefficient is our objective unction in this work and we try to maximize this function to get he maximum sound energy transmission to the air media by ZT transducer. Therefore best solutions are those chromosomes hat maximize the objective function. But this maximum value is elative. Therefore we need a stop criterion to terminate the algo-ithm. For preventing of local solution we used mutation method o generate 5% of parents and the rest of them generate by One oint Crossover method. For present work, we consider 0.7 values s a stop criterion. If the objective function is less than stop crite-ion the algorithm must generate new chromosome. Each material coustic impedance converts to binary code and by One Point rossover method new chromosome is generated and again new ransmission coefficient (objective function) is calculated and com-ared with stop criterion (0.7). This process continues to find some ice and suitable chromosome.

.2.2. Explanation of written program

In this work, according to GA a program was created by Matlab oftware. The schematic is illustrated in Fig. 4. In this program, a set of acoustic impedances as initial population due to number of layers was generated randomly. These values must be between PZT ($Z_p = 3.3 \times 107$ kg/(m^2 s)) and air ($Z_a = 427$ kg/(m^2 s)). After that the transmission coefficient was found with these acoustic impedances and the result was compared with the stop criteria (0.7). if this value is less than stop value, this chromosomes change by One Point Crossover method and the calculation of transmission coefficient repeat again and the result check with stop value and so on. This procedure has done until the transmission coefficient be bigger than stop value, in this state these set of chromosomes (acoustic impedances) accepted as a goal. This program can be run well for every number of layers. For summary, basically the genetic algorithm performs the following loop:

1. randomly create an initial population (a set of acoustic impedances is generated as chromosome),
2. repeat,
3. valuate the fitness of each individual,
4. select one or more individuals from the population with a probability based on fitness to participate in genetic operations,
5. create new individuals by applying genetic operations with specified probabilities,
6. until an acceptable solution is found or some other stopping condition is met (transmission coefficient = 0.7),
7. return the best-so-far individual.

Gene provides the implementation of the common genetic operator that called "One Point Crossover". In this step new chromosome chooses randomly one crossover point from parent chro-

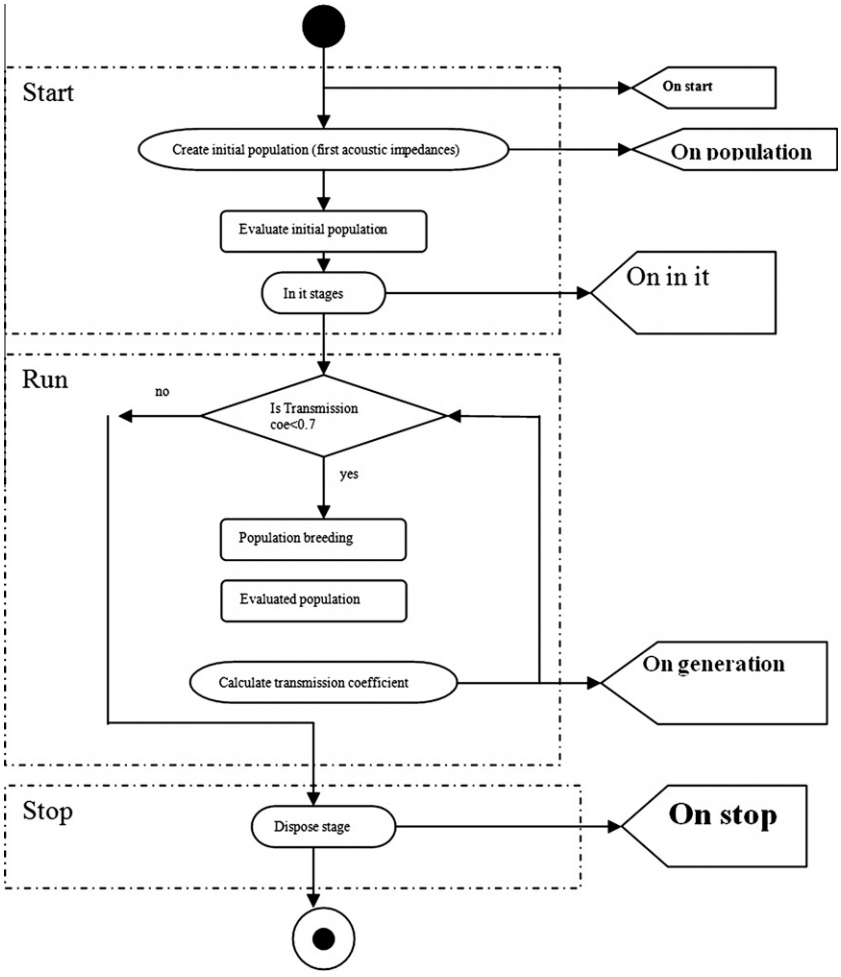


Fig. 4. Schematic algorithm that is used in present work.

Table 3
Chromosomes and offspring relation.

Chromosome 1	11101|001000
Chromosome 2	00001|010101
Offspring 1	11101|010101
Offspring 2	00001|001000

mosomes and creates a new offspring. One point crossover can look like Table 3. (where "|" is the crossover point).

In this case we used "simple mutator method" to modification. The simple mutator chooses randomly a mutation point and randomizes its gene. Mutation operator also needs a probability number which specifies its use. Generally for a mutation operator the probability is set to 20%. Jens provides a flexible structure for evolving a population of individuals. The used genetic algorithm is structured as depicted in Fig. 4.

The algorithm execution passes through the following events:

- Start: the algorithm evolution starts. The random acoustic impedances are generated according to number of layers
- Init: internal structures, such as the population given as input at generation 0 and history, are initialized.
- Generation: a generation has been just performed.
- Stop: the algorithm terminates its executions. The stop criterion is amount of transmission coefficient and when it is more than 0.7 the process is terminated.

4.3. Results

4.3.1. Acoustic impedance determination

To design a transducer it is enough to model the energy sourc such as PZT and matching layers by adding a backing layer in Aba qus software to find the longitudinal resonance frequency b changing backing and PZT and a bit near matching layer(s) thick nesses. for instance, Fig. 5 shows the PZT transducer for four laye works at resonance frequency after matching calculating and natu ral frequency analysis. Therefore, for the first step, we need to cal culate the matching layers. In this work, we only focus t calculating matching layer(s). For do that, we consider PZT and ai as medias and concentrate to calculate the layers between them By applying the above genetic algorithm (Section 4.2), we com puted the acoustic impedances of matching layers for a total num ber of media (n) or ($n-2$) number of layers ranging from 3 to 8 i frequency range of 0.02–2.5 MHz. The values of the optimal acous tic impedances are given in Table 4. Besides, we depict in Fig. 5 th corresponding transmission coefficients with a dotted gray line. I the same figure, we plot, the two curves corresponding to th Chebyshev theory (black solid line). The transmission coefficient obtained with the genetic algorithm are also given in Table 4.

The optimal transmission coefficient obtained with geneti algorithm is 0.8625 who correspond to the 6th layers (Table 6 The materials for layers from direction of PZT to air are: Lead meta niobate, Araldite-90phe, Fused silica and PTFE. It is not mean tha this set is the best one in all criteria such as manufacturing.

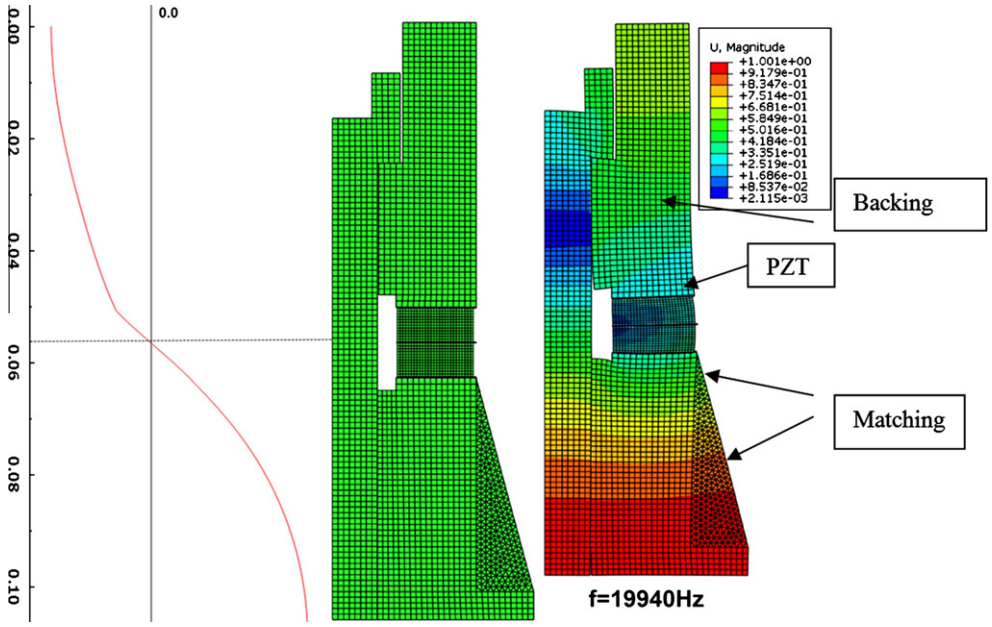


Fig. 5. Finding the longitudinal resonance mode of the transducer in Abaqus.

Table 4
Acoustic impedances obtained by a genetic algorithm.

Number of media n	Acoustic impedance (1000 kg/(m^2 s))									Transmission coefficient
	Layer 1	Media 2 to ($n-1$)							Media	
3	35,000	131							0.427	0.7391
4	33,000	26,400				131			0.427	0.7195
5	33,000	27,300			4050		17		0.427	0.7582
6	33,000	9020		1600			990	17	0.427	0.8625
7	33,000	14,090	11,880	9020				2000 17	0.427	0.8295
8	33,000	30,100		26,400	13,000	6650	1710	17	0.427	0.8349

Table 5
Sub-optimal solutions ($n = 3$).

Acoustic impedance of second layer (1000 Kg/(m^2 s))	Transmission coefficient
190	0.7921
131	0.7391
81	0.6359
100	0.6031

able 6
ub-optimal solutions ($n = 4$).

Acoustic impedance of second layer (1000 kg/(m^2 s))	Acoustic impedance of third layer (1000 kg/(m^2 s))	Transmission coefficient
30,100	190	0.7233
26,400	131	0.7195
26,400	109	0.6679
18,700	100	0.6558

able 7
ub-optimal solutions ($n = 6$).

Acoustic impedance of second layer (1000 kg/(m^2 s))	Acoustic impedance of third layer (1000 kg/(m^2 s))	Acoustic impedance of forth layer (1000 kg/(m^2 s))	Acoustic impedance of fifth layer (1000 kg/(m^2 s))	Transmission coefficient (1000 kg/(m^2 s))
20,500	16,000	12,550	81	0.7022
31,000	17,330	3860	17	0.7103
21,500	12,810	12,550	17	0.8636

means that this is a good set for high transmission energy to the air load. But existence of a lot of set of layers by using GA is important because the flexibility can be increase. This is also the case of transmission coefficients corresponding to the exact values of acoustic impedances obtained by the Chebyshev theory. However, these theoretical acoustic impedances do not correspond systematically to an available material. In order to achieve the theoretical transmission coefficients new materials should be designed which is far to be simple task. Therefore, the genetic algorithm has the significant advantage to achieve very high transmission coefficients with available materials.

Furthermore, the genetic algorithm yields sub-optimal solutions with very good transmission coefficient and only by using the materials available in the database and not new ones. This is important if any material is expensive or there is shortage in supply. In Tables 5 and 6, we give sub-optimal solutions in the case of ($n = 3$) and ($n = 4$), respectively. The same task can be achieved for ($n > 4$). In the case of ($n = 3$), we have four sub-optimal solutions with a transmission coefficient higher than 0.6 (see Table 5). Besides, we have three sub-optimal solutions with a transmission coefficient higher than 0.7 in the case of six layers (see Table 7). As illustrated in Table 5 the transmission coefficient for $Z = 131{,}000$ kg/(m^2 s) correspond poly ether sulfone is a bit less than $Z = 190000$ kg/(m^2 s) correspond to hydrogen, solid at 4.2°K. But it is more a lot practical.

4.3.2. Effect of thickness variation on frequency

Both the variation of frequency by changing thickness and variation of transmission coefficient in constant frequency carried out. For knowing the effect of frequency, we consider two working frequencies, the first is low (50 kHz) and the other is high (2 MHz) and for both frequency, the specification of thickness and frequency of $n = 4$ and 6 layers are presented in Table 8.

able 8
hickness characteristic for two number of layers.

						Transmission coefficient
Layer number						
n = 4						
Acoustic impedance of second layer (1000 kg/(m^2 s))		26400		131		
Thickness (mm)	Frequency 50 kHz	55		2.24		0.7195
	2 MHz	1.4		0.056		
n = 6						
Acoustic impedance of second layer (1000 kg/(m^2 s))		21500	12810	12550	17	
Thickness (mm)	Frequency 50 kHz	11	7.6	28.5	4	0.8636
	2 MHz	0.275	0.19	0.7125	0.1	

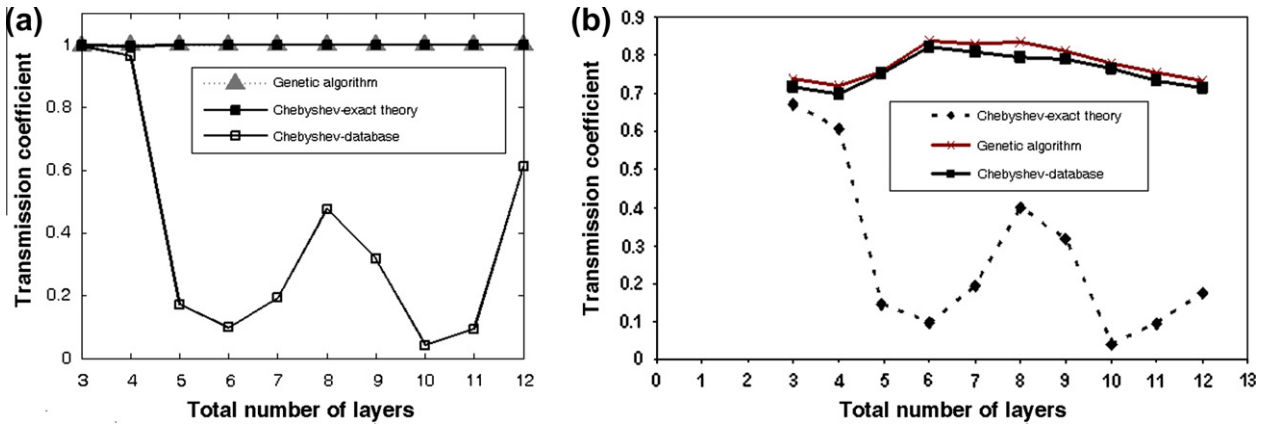


Fig. 6. Transmission coefficients obtained by a genetic algorithm.

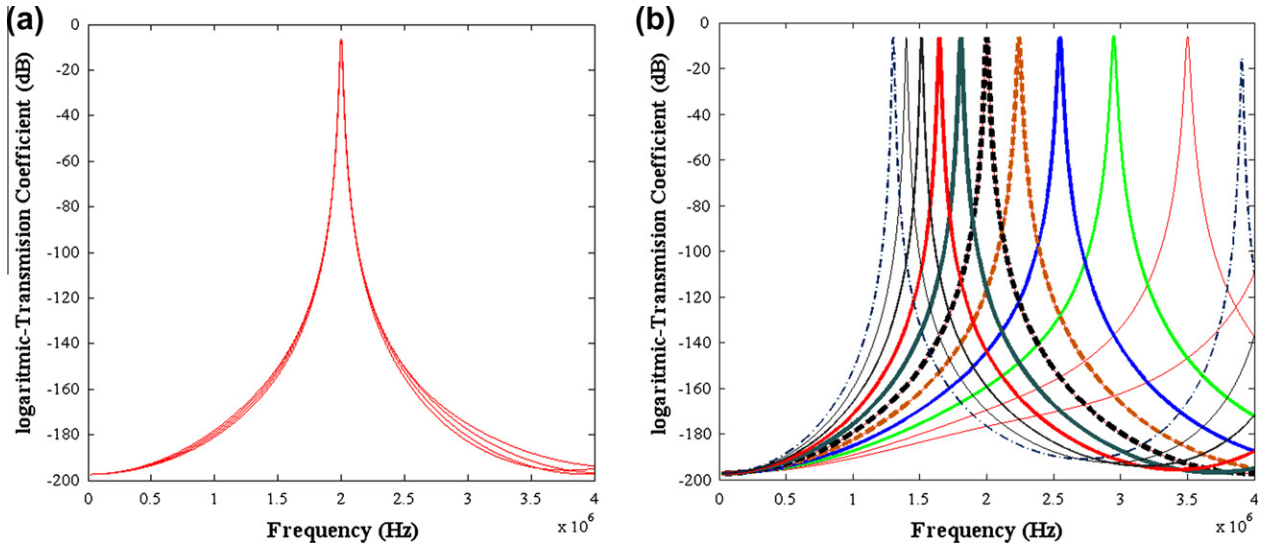


Fig. 7. Frequency dependency to the thickness variation for $n = 4$ and 2 MHz (a) second layer and (b) for third layer.

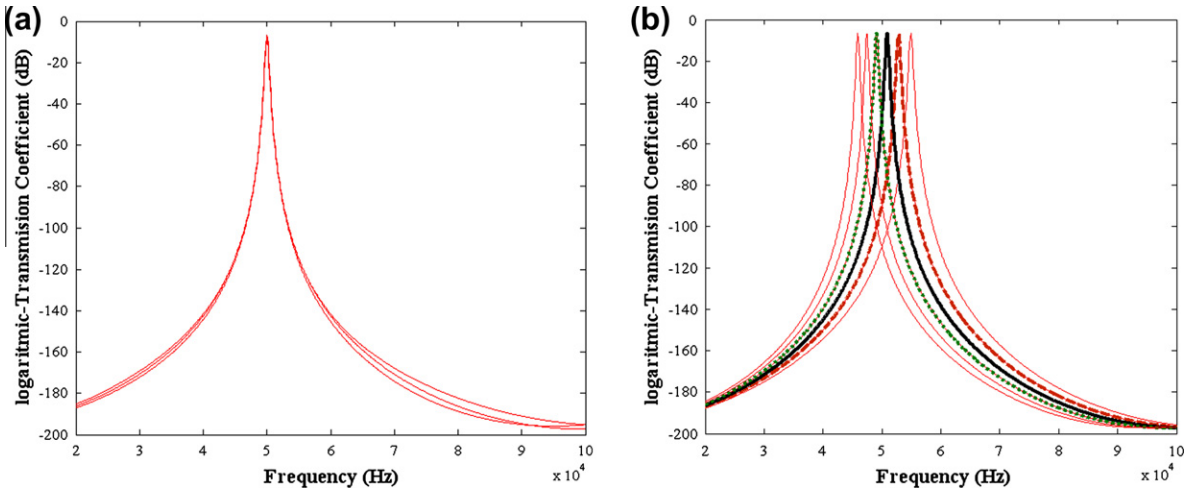


Fig. 8. Frequency dependency to the thickness variation for $n = 4$ and 50 kHz (a) second layer and (b) for third layer.

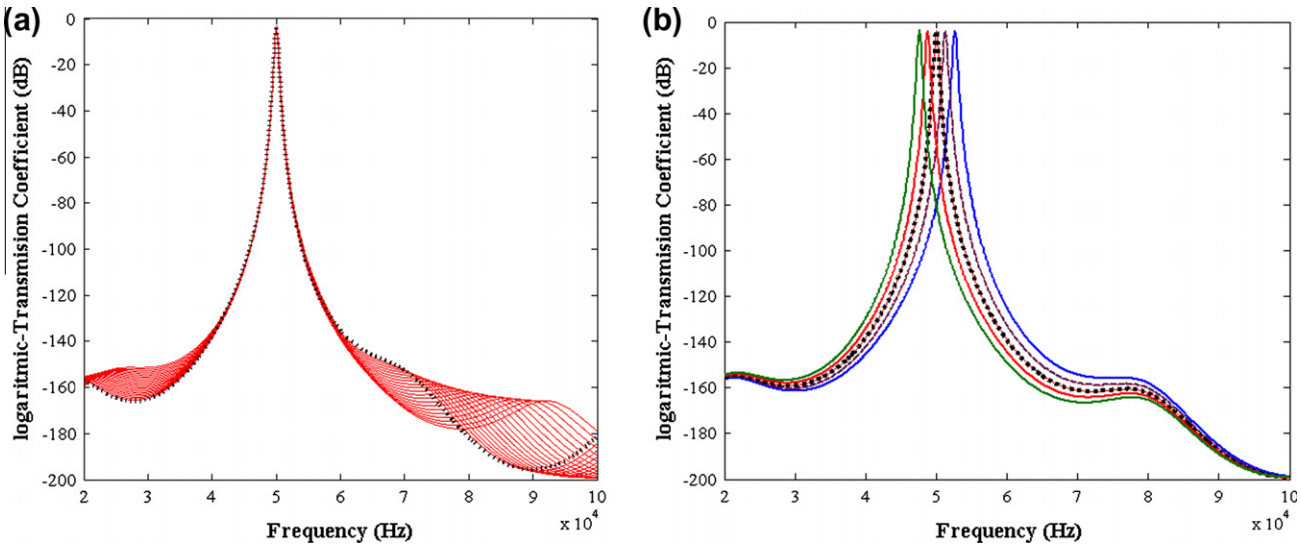


Fig. 9. Frequency dependency to the thickness variation for $n = 6$ and 50 kHz (a) forth layer and (b) for fifth layer.

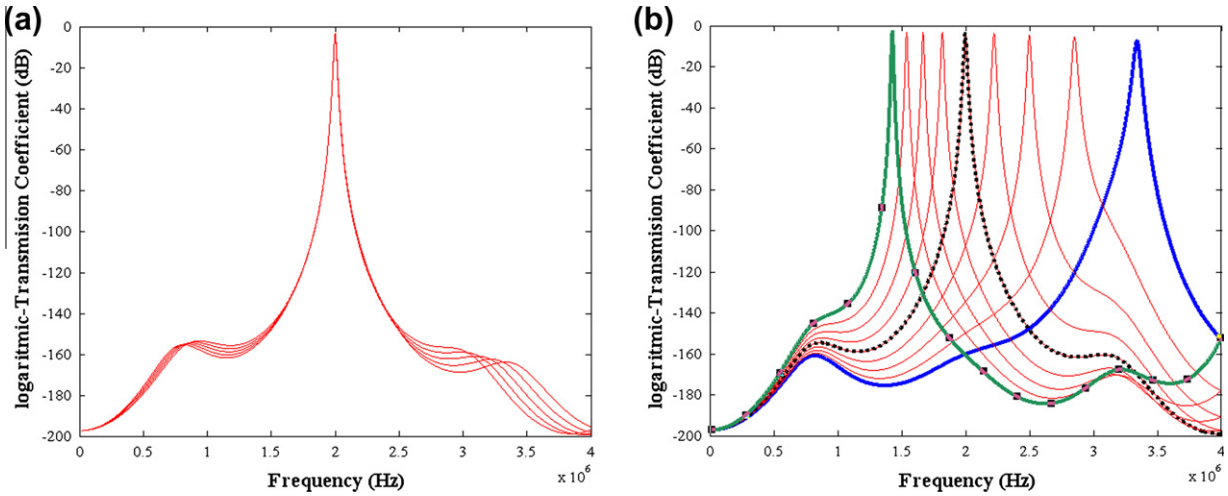


Fig. 10. Frequency dependency to the thickness variation for $n = 6$ and 2 MHz (a) forth layer and (b) for fifth layer.

able 9
ayer thickness for different frequency for 1 layer.

Layer number = 1	*z*=33,000,000	131,000	427						
Frequency	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000	100,000
Layer thickness (mm)	0.01875	0.0125	0.009375	0.0075	0.00625	0.005357	0.004688	0.004167	0.00375
Transmission coefficient	0.996	0.996	0 996	0.996	0.996	0.996	0.996	0.996	0.996

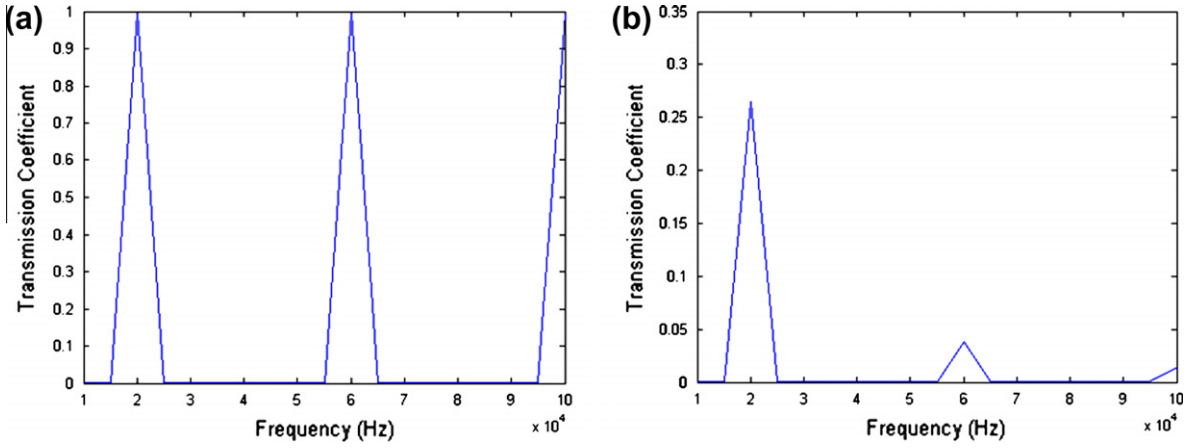


Fig. 11. Calculated transmission coefficient for 1 layer (a) for 0.01875 m layer thickness and (b) for 0.0187 m layer thickness.

.3.2.1. Transmission coefficient sensitivity to frequency variation. By ssuming the ±40 μm tolerance of thickness for the layer who is onnected directly to the air (last layer in two cases), the frequency hift between 1.4 and 3.3 MHz for 5th layer of six layers while at he same situation for 4 layer frequency band is between 1.2 and .5 MHz (Figs. 7b and 10b). It means that by increasing the number f layer the sensitivity to the frequency shift decrease at the same ariation in the thickness. On the other hand, the pervious layers or both cases are not sensitive to the thickness variation (Figs. a and 10b). The same strategy has done for both numbers of lay-rs in low frequency (50 kHz). We use ±0.2 mm as thickness band ariation to find out the influence of that on frequency shifting. The esults in Figs. 8b and 9b illustrate the some more sensitivity in last ayer of four layers matching system rather than 6 matching layers. ince the frequencies locate in domain of [45, 56 kHz] in four ayers. Meanwhile, the same parameter is [48, 54 kHz] for last layer f six matching layers. By Comparison the low and high frequency, s found out the sensitivity of frequency shifting to thickness

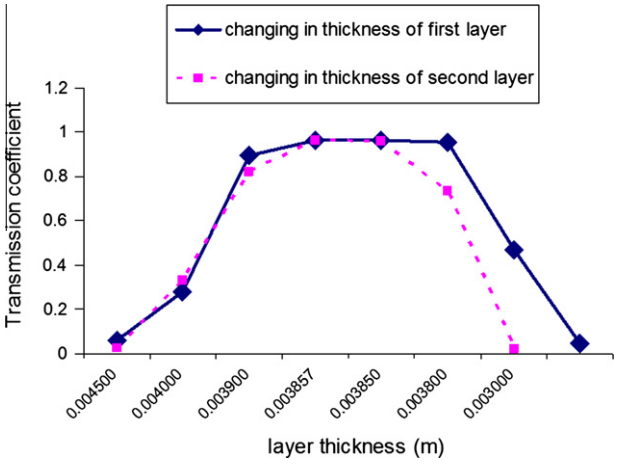


Fig. 12. Transmission coefficient sensitivity to thickness variation for 2 layers.

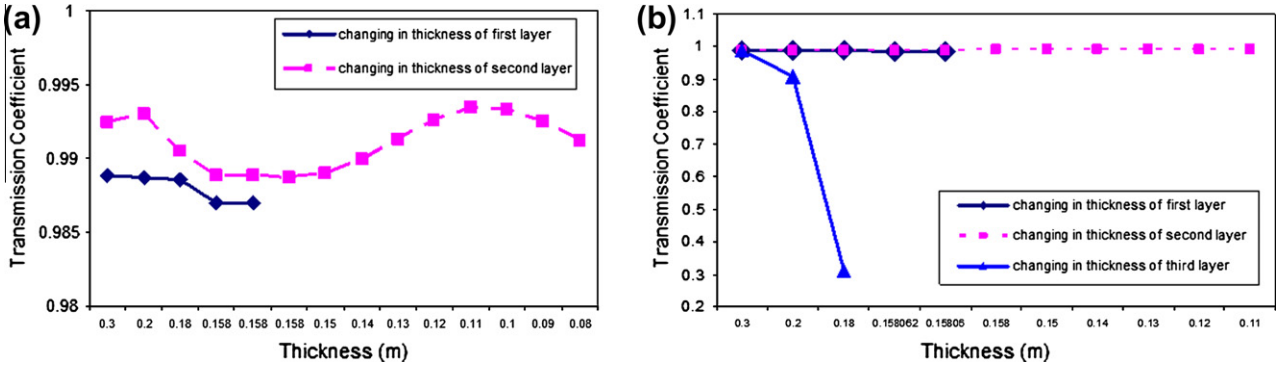


Fig. 13. Transmission coefficient sensitivity to thickness variation for 3 layers (a) first and second (2 layers) and (b) for all layer (3 layers).

variation is more important in high frequency. Again in low frequency, in previous layers there is no significant sensitivity for frequency variation due to the thickness variation (see Figs. 8a and 9a).

4.3.2.2. Transmission coefficient sensitivity to thickness variation. Another important issue is the role of thickness on transmission coefficient. Especially for transducer that works in unique or single frequency. In this stage, for both preventing of irritation similarities and considering only the effect of thickness variation on transmission coefficient (at low frequency the attenuation coefficient effect is less than high frequency), we only concentrat on low frequency domain and let us to assume there is no attenu ation in this section. For investigating on layer thickness effect o transmission coefficient and its sensitivity is enough to chang layer thickness and calculate transmission coefficient for eac thickness. For finding this effect, we assume all layer's thicknes are calculated according to $(2m-1)\pi\frac{\lambda}{4}$ equation (λ is wave length for example for 3 medias or 1 layer (PZT, polyethersulfone and air the thickness of polyethersulfone layer change from 0.01875 r (the thickness for 22 kHz) to 0.0187 m. there are some calculate thickness for different frequency in Table 9. As shown in Fig. 11

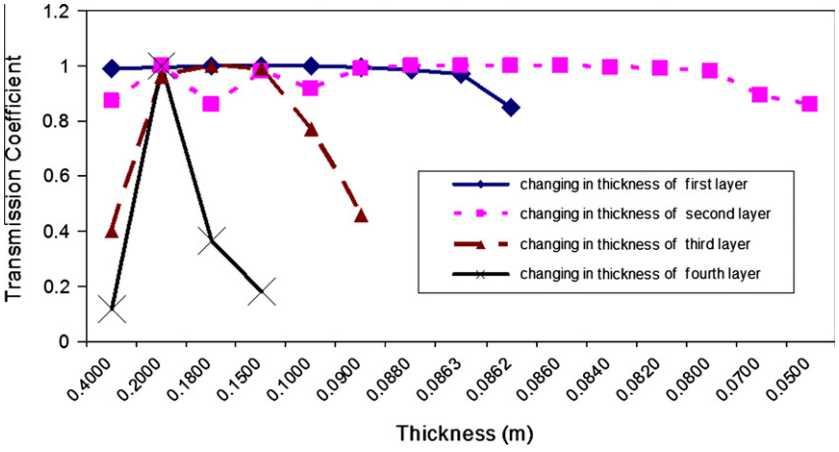


Fig. 14. Transmission coefficient sensitivity to thickness variation for 4 layers.

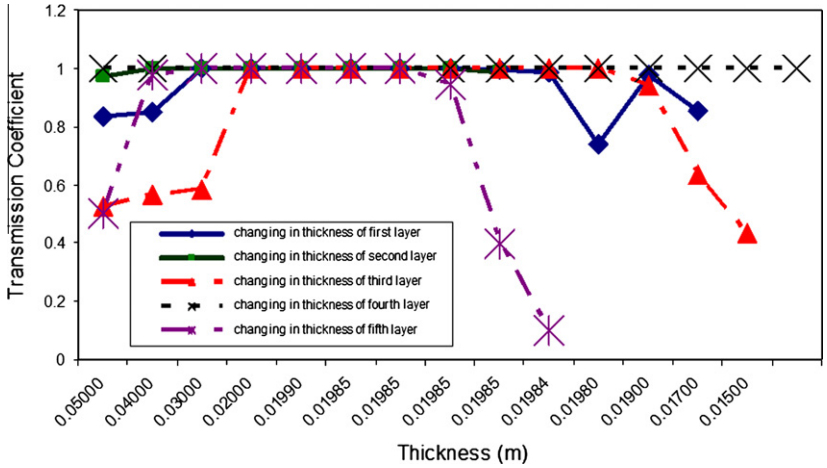


Fig. 15. Transmission coefficient sensitivity to thickness variation for 5 layers.

able 10
alculated thickness from $(2m-1)\pi\frac{\lambda}{4}$ theory.

Number of layer = l				Transmission coefficient
Z(kg/(m^2 s))	33,000,000	131,000	427	0.9966
$L(m)$		0.0046875		

able 11
alculated thickness from GA theory.

Number of layer = l				Transmission coefficient
Z(kg/(m^2 s))	33,000,000	100,000	427	0.963134951
$L(m)$		0.779997466		
Z	33,000,000	131,000	427	0.92426589
L		0.99843960S		
Z	33,000,000	98,000	427	0.977369947
L		0.577499657		
Z	33,000,000	150,000	427	0.918201609
L		0.980693103		

Table 12
Transmission coefficient calculated by GA for different thickness.

Thickness (m)	Transmission coefficient
0.9986	0.0102
0.99855	0.0251
0.99848	0.0303
0.9984756	0.0374
0.9984542	0.168
0.9984454	0.4737
0.99844	0.8974
0.9984385	0.9793
0.9984375	*0.9966*
0.998439603	0.92426589
0.998435	0.8974
0.998425	0.2648
0.998415	0.1001

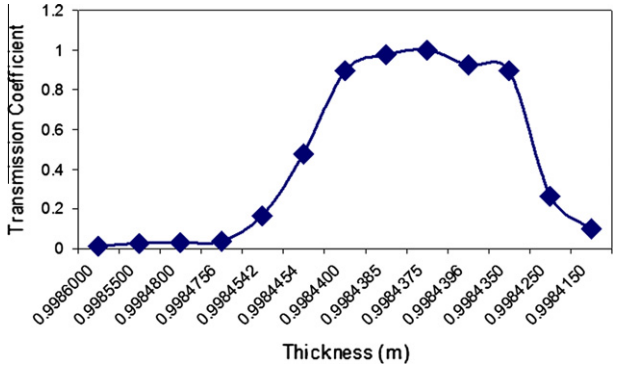


Fig. 16. Transmission coefficient versus thickness (m).

he transmission coefficient decrease from 0.986 to 0.265. It shows high sensitivity to thickness variation. In the next trial, the thickess changes to 0.018 m. This result expresses a very important act that manufacturing accuracy should be very high and thickess tolerances must not bigger than ± 10^{-6} m (1 μm). In 4 media r 2 layers at the first step for understanding of transmission oefficient sensitivity to thickness variation, we change the first ayer thickness. In this state the thickness of second layer hold contant. For determination of effect of second layer on transmission

Table 13
Thickness calculated according to $(2m-1)\pi\frac{\lambda}{4}$ and GA by choosing acoustic impedances from GA for 2 layers.

Acoustic impedance (z) (kg/(m^2 s))	33,000,000	8,650,000	63,000	427
Layer thickness according to $(2m-1)\pi\frac{\lambda}{4}$ (m)		0.014719	0.004125	
Layer thickness according to GA (m)		0.20952	0.939116832	

Table 14
Thickness calculated according to $(2m-1)\pi\frac{\lambda}{4}$ and GA by choosing acoustic impedances from GA for 3 layers.

Acoustic impedance (z) (kg/(m^2 s))	33,000,000	21,500,000	2,160,000	63,000	427
Layer thickness according to (2m-1)$\pi\frac{\lambda}{4}$(m)		0.036667	0.032667	0.007333	
Layer thickness according to GA (m)		0.0286628	0.659882	0.217492528	

coefficient it is enough to change the second layer thickness while the first layer is constant. The results are illustrated in Fig. 12. This figure illustrate that transmission coefficient is much sensitive to second layer rather than first one.

For investigating the influence of layer thickness on transmission coefficient, the best set of acoustic impedances that determined by GA, is selected for each number of layers (3, 4 and 5 layers). Now by fixing layer thickness of all layers except the thickness of first layer the transmission coefficient is calculated. After that except second layer, the rest layers hold constant and thickness of second layer is changed. This process is repeated for other layers. The results are presented in Figs. 13–15.

According to the above results, it is found out that with increasing of number of layers the sensitivity of layers to variation of thickness is decreased. There for according to manufacturing, it is better to choose more number of layers to decrease the manufacturing limitation. But by increasing the number of layers the efficiency decrease. Then, finding an optimum number of layers is very important.

4.3.2.3. Comparison between thicknesses calculated from GA and equation $(2m-1)\pi\frac{\lambda}{4}$ for different number of layers. Maybe this question is asked "is $(2m-1)\pi\frac{\lambda}{4}$ formula optimum for layer thickness?" for answering to this question genetic algorithm is used and the results from that compromise with results based on the $(2m-1)\pi\frac{\lambda}{4}$ equation. The GA used in this work able to calculate the best thickness and acoustic impedance to obtain highest transmission coefficient. Therefore, we can say, if the obtained transmission coefficient from GA to be more than one obtained from $(2m-1)\pi\frac{\lambda}{4}$, then $(2m-1)\pi\frac{\lambda}{4}$ is not the optimum method to determination of layer thickness for maximum transmission coefficient. First step is started with 3 media or 1 layer. The results showed that the thickness is 0.998439603 with acoustic impedance equal to 131,000 kg/(m^2 s) and frequency equal to 22 kHz (Tables 10 and 11). By a little attention to this value, it is understood this value is very close to 213 ∗ 0.0046875 (the value calculate by $(2m-1)\pi\frac{\lambda}{4}$). Therefore if 0.998439603 to be highest value for thickness layer that calculated from GA theory, it is prove that

Table 15
Thickness calculated according to $(2m-1)\pi\frac{z}{4}$ and GA by choosing acoustic impedances from GA for 4 layers.

Acoustic impedance (z) kg/(m^2 s)	33,000,000	24,100,000	23,200,000	8,250,000	63,000	427
Layer thickness according to $(2m-1)\pi\frac{z}{4}$		0.161125	0.02875	0.03225	0.0055	
Layer thickness according to GA (m)		0.286646	0.54404	0.974233	0.273156828	

Table 16
Comparison between GA and $(2m-1)\pi\frac{z}{4}$ for different number of layers.

Layered number	Frequency (kHz)	Transmission coefficient	
		GA	$(2m-1)\pi\frac{z}{4}$
4	80	0.992170061	0.6672
5	45	0.991853533	0.1222
6	60	0.996994802	0.6601

the thickness that calculated from $(2m-1)\pi\frac{z}{4}$ is the optimum one. To understand this fact, the transmission coefficient are calculated for some thicknesses around 0.998439603 (Table 12 and Fig. 16). As shown in Table 12 and Fig. 16, it specifies that the highest transmission coefficient occur at 0.9984375 m that is exactly equal to $213 * 0.0046875$. Therefore we can say the best theory for calculating thickness for 1 layer is $(2m-1)\pi\frac{z}{4}$ Fig. 6.

For understanding which theory is better for 2, 3 and 4 layers, it is enough to generate some thickness and acoustic impedance by GA and for one of them, calculate thickness based on $(2m-1)\pi\frac{z}{4}$ with acoustic impedances obtained from GA. Now if the calculated transmission coefficient is higher than one calculated from GA, an then $l=(2m-1)\pi\frac{z}{4}$ is optimum. Other wise the GA is optimur and we can say that for 2, 3 and 4 layers $(2m-1)\pi\frac{z}{4}$ is not bes solution but it may be a good solution for choosing thickness lay ers. The results get from GA for 2, 3 and 4 layers are presented i Tables 13–16 and Figs. 17–19. For instance, as illustrate i Fig. 12, the maximum transmission coefficient occur at 80 kH for 4 layers. For finding thickness according to $(2m-1)\pi\frac{z}{4}$, acousti impedances values that generate value of 0.992170061 is used fo four layers. The calculated thicknesses with these two methods ar shown in Table 13. Since the maximum transmission coefficien occurs at 80 kHz in GA method, it must choose thicknesses a 80 kHz and calculate transmission coefficient for it. The resul illustrated in Fig. 17. By comparison between Figs. 17a and b i understood the transmission coefficient obtained from GA (calcu lated transmission coefficient equal to 0.992170061) is higher tha one gets from $(2m-1)\pi\frac{z}{4}$ (calculated transmission coefficien equal to 0.6672). It means thickness from $(2m-1)\pi\frac{z}{4}$ is not th optimum method for calculating layers thicknesses in four layer number. The same results obtained for 5 and 6 layers. The result are presented in Figs. 18 and 19 and Tables 14–16.

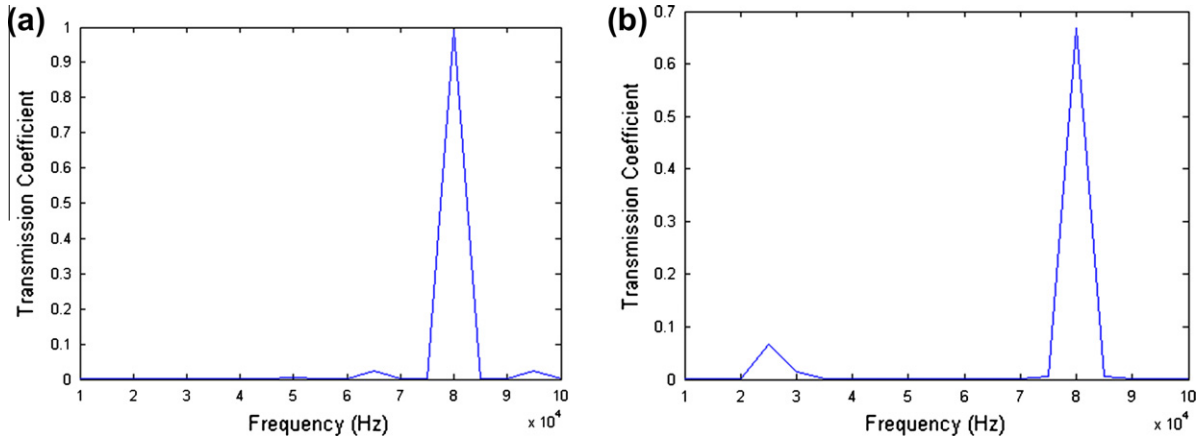


Fig. 17. Calculated transmission coefficient at different frequency for 2 layers ((a) according to GA and (b) according to $(2m-1)\pi\frac{z}{4}$).

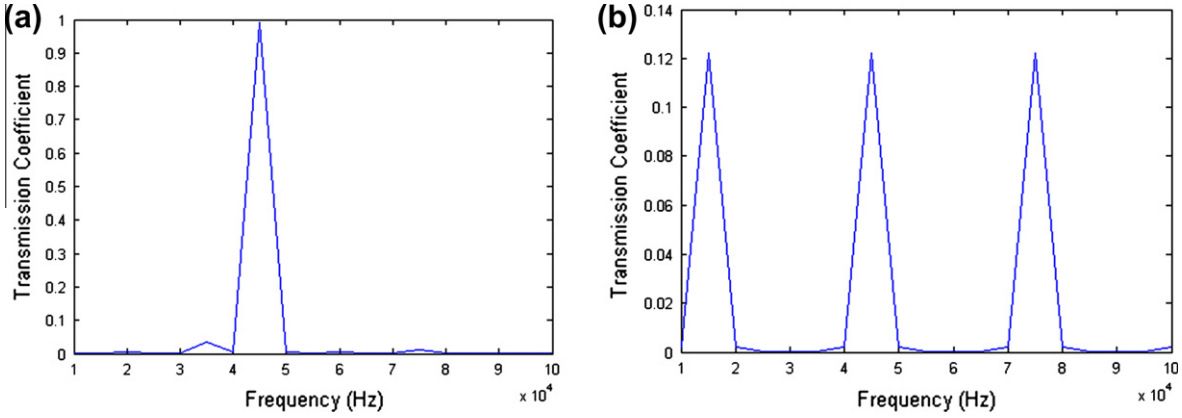


Fig. 18. Calculated transmission coefficient at different frequency for 3 layers ((a) according to GA and (b) according to $(2m-1)\pi\frac{z}{4}$).

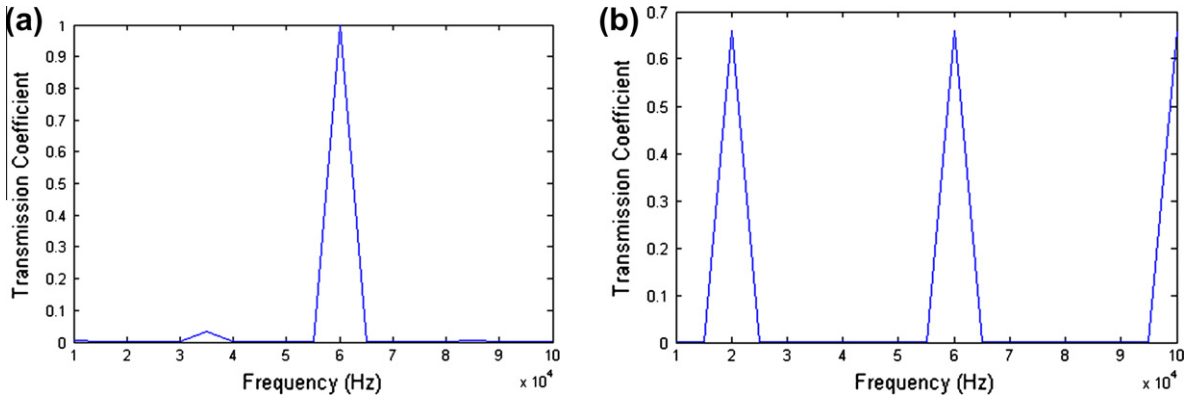


Fig. 19. Calculated transmission coefficient at different frequency for 4 layers ((a) according to GA and (b) according to $(2m-1)\pi\frac{\lambda}{4}$).

According to above results, it is found out that the $(2m-1)\pi\frac{\lambda}{4}$ quation just convenient and optimum for calculating layer thickess for 1 layers and not optimum for more than 1 layer.

. Conclusion

In this paper, we are interested in the calculation of the acoustic npedances of matching layers and thickness of layers for ultraonic airborne transducers. Four methods are presented: three nalytical (Chebyshev, Desilets and Souquet) and one numerical using a genetic algorithm). The acoustic impedances are calcuated for sake of optimal transmission coefficient. This paper has resented a comparison between layer thicknesses that specifies ccording to ideal quarter-wave layers and specify by GA method nd determination of sensitivity of transmission coefficient to layrs thickness.

The advantage of this method lies in the use of simple materials o make a matched transducer. The performances of the proposed natched transducer are similar to those obtained with the ideal ne quarter-wave layer matching. Once materials that constitute he two matching layers are selected, only their thicknesses have o be calculated.

According to the above results, we can make the following onclusions:

- The Chebyshev theory gives the best theoretical transmission coefficient for the all studied cases.
- The acoustic impedances obtained by the three theories do not systematically correspond to readily available materials. As elaborating new materials is a hard and expensive task, the theoretical acoustic impedances are switched to the nearest value available in a wide material database. This induces a sharp drop in the transmission coefficients for a number of layers higher than or equal to four.
- The genetic algorithm solutions have the main advantage of corresponding to nowadays available materials. Moreover, they yield transmission coefficients almost equal to unity.
- For sake of flexibility, we can get, with genetic algorithm, multiple sub-optimal solutions which have very good transmission coefficient for each case (fixed number of layers).
- Sensitivity of frequency shifting to the thickness variation, increase by increasing the frequency and this sensitivity decrease by using more layers matching
- There is no significant sensitivity to frequency shift in layers far away from the air.
- Transmission coefficient strongly depends on layers thickness in air born ultrasonic transducer.
- Sensitivity of transmission coefficient to thickness decrease with increasing of number of layers.
- The latest layer (the layer is next to air) plays the basic and important role to value of transmission coefficient.
- The Chebyshev theory is the best choice for 1 layer both determines acoustic impedance and layer thickness.
- The $(2m-1)\pi\frac{\lambda}{4}$ equation is not the optimum method to determine layer thickness to maximize the transmission coefficient for more than 1 layer.

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